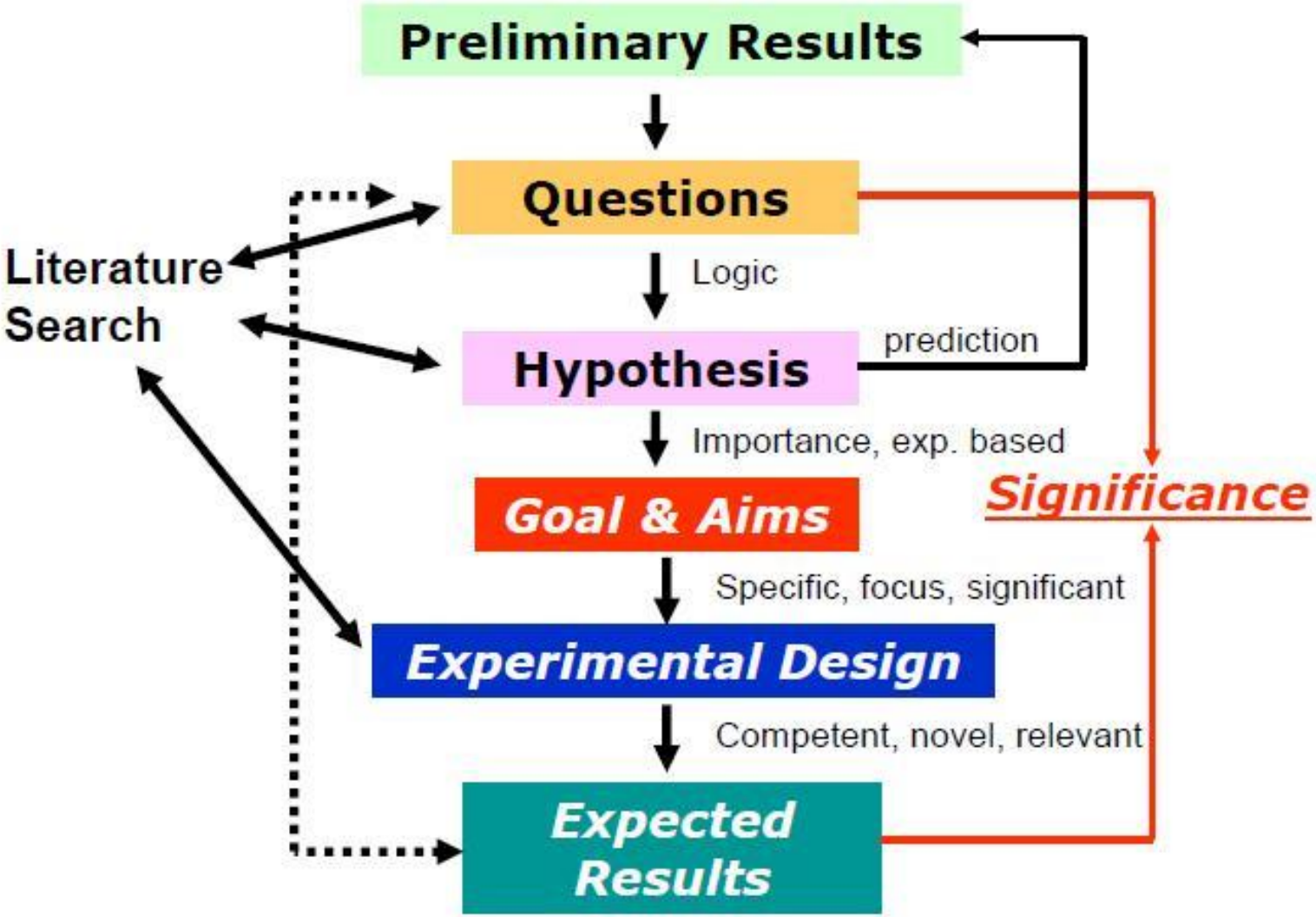
Writing Research Proposal



Benefits of Writing Proposal

- Fund to support your research
- Allow you to review and critically evaluate your current research program
- Convince yourself and others that your research is worth supporting
- Keep you focused on your research program
- Develop novel ideas during writing proposal


Preliminary Results
Questions
Literature Search
Logic
Hypothesis
prediction
Importance, exp. based
Goal & Aims
Significance
Specific, focus, significant
Experimental Design
Competent, novel, relevant
Expected Results

Key Points of Your Proposal

***What** are your objectives*

***Why** is your study important*

***How** are you going to do it*

Keys to good proposal writing

Clarity, Understandable

Evidence-based

Conciseness

Relevancy; to the point

Logic

Keys to writing a successful proposal

- Interesting and/or important problem
- Clear Concept and Rationale; ***Innovation***
- Reasonable and achievable aims
- Easy to understand
- Appropriate experimental designs with feasible and up-to-date techniques
- Interesting preliminary/progress results and previous publications relevant to this proposal
- **Competence of PI and record of publication**

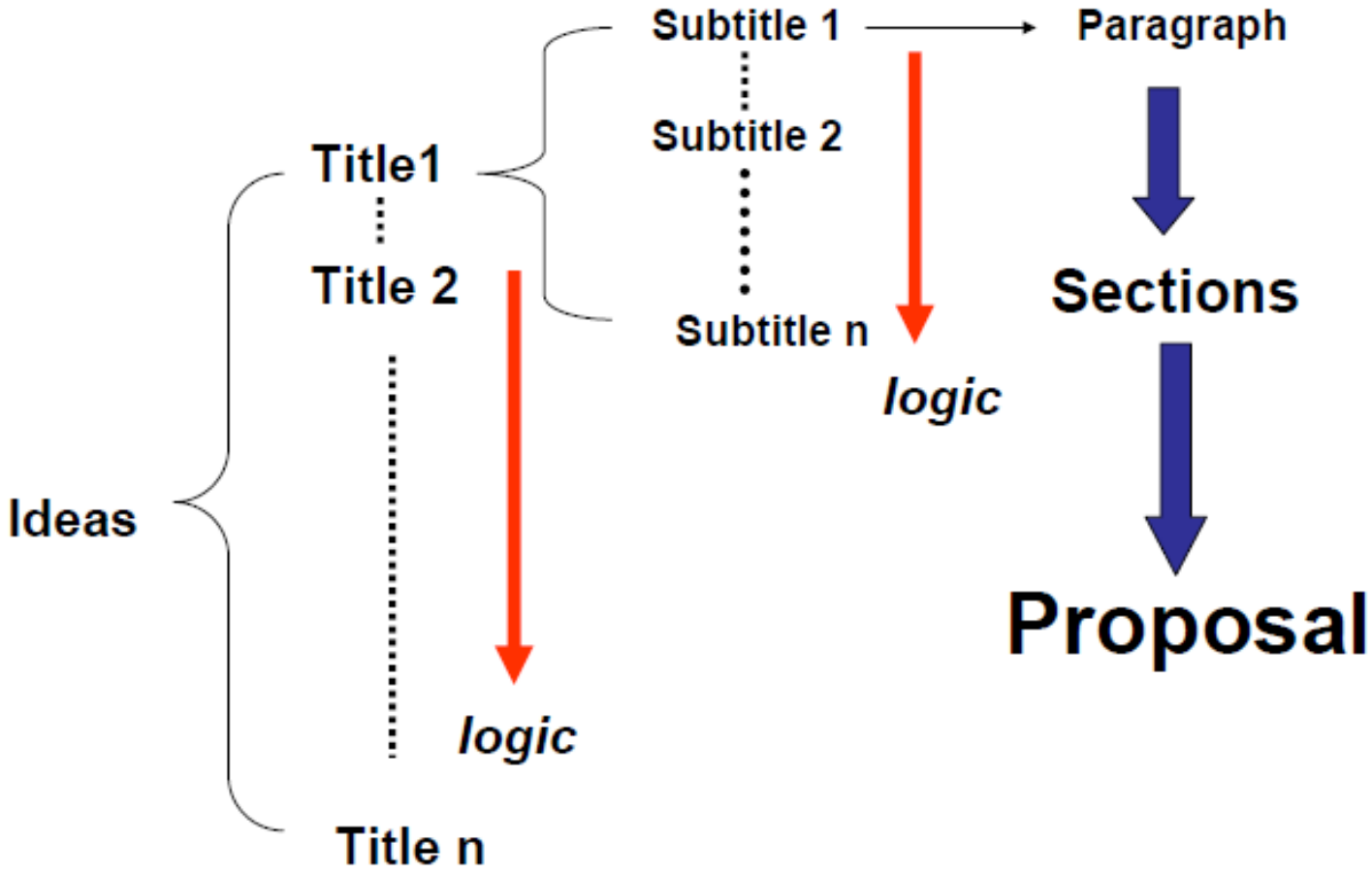
How to start writing proposal(1)

- Write Only **Outline** at the first draft; do not start by writing full proposal

- Write **Abstract** at the end

- Write **Title** of Proposal when the proposal is finished

Writing technique: from ideas to paragraphs to sections to full proposal



Writing a Proposal(1)

- Title(**8**)
- Abstract(**7**)
- Background and Rationale(**2**)
- Overall Goal and Specific Aims(**1**)
- Significance(**3**)
- Preliminary Results (*relevant ones)*(***4***)
- Experimental Design(**5**)
- Expected Results(**6**)
- Budget(**9**)
- ***Check list (for your own use)***

Writing a Proposal(2)

•Ideas, Ideas, Ideas! Write them done any time in a file or a small notebook!

•Based on *facts ; no speculations*

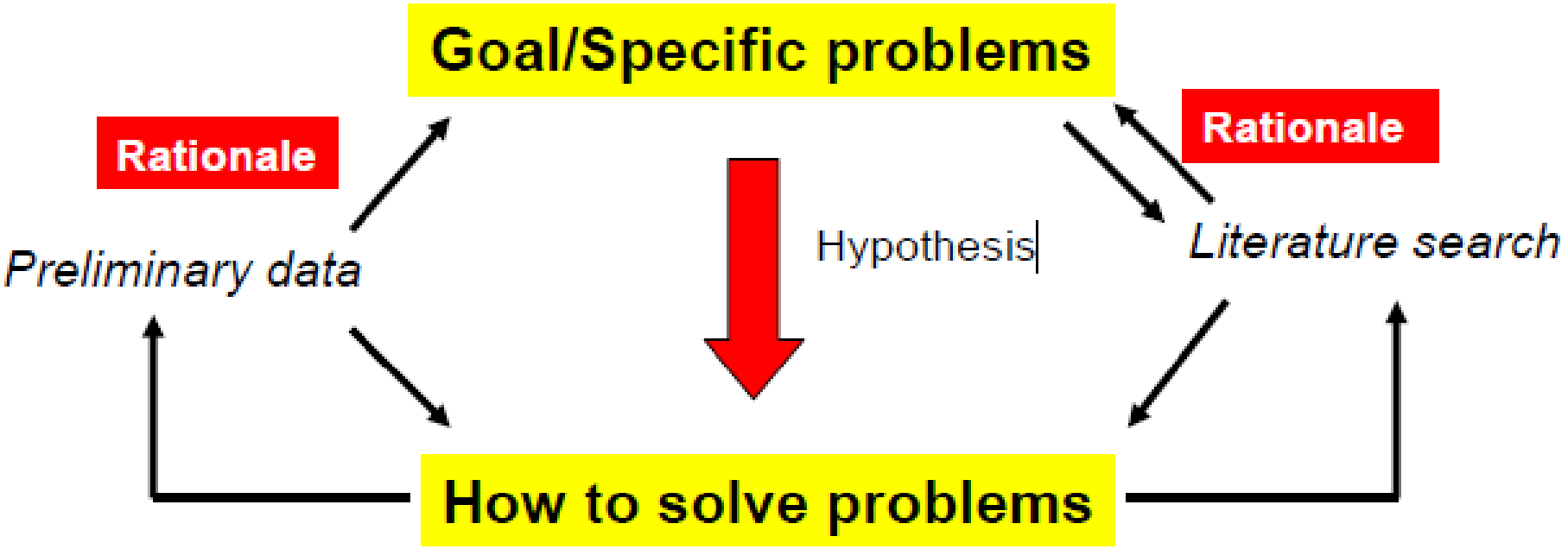
•Checking in literatures for similar ideas (usually in Introduction and Discussions) done in other systems---Key words

•*Evolution of ideas* ⇨ *Hypothesis*

•Obtain preliminary results to support your Hypothesis

Initial Planning: start early!

How to start(2): how to get ideas



Introduction

- **Previous studies**: *General background information; your contribution*
- **Important and interesting problems still outstanding**: *relevant to this work: why it is important; selling points*
- **How the problem will be solved**: *hypothesis and experimental approaches*
- **Summary and significance**

Introduction: *Problems*

- Too little or irrelevant background information; **information not updated**
- Unclear purpose; **no rationale for the study**
- Disorganization in logic

Hypothesis

- Build the case to support the hypothesis with literature and data---***no reinventing the wheel***
- ***Seeing is believing***: Use schematic figures or diagrams to help reviewers understand your model and idea
- State your hypothesis clearly in the abstract and specific aims; alternative hypothesis in case

Hypothesis: *Problems*

No hypothesis; Fishing Expedition

Wild speculation

Too complicated, hard to understand

Preliminary Results/ Progress Report

- Show ***only*** the relevant experiments ***supporting*** your hypothesis
- The results should not be ambiguous
- Figure and table legends should be clearly written; figure numbers should correspond to the text; be sure to label the figures
- Do not show published results in this section

Figures and Tables

- Simple and Clarity in presentation
- Visualization presentation techniques
- Caption must stand-alone
- Error bars
- Resolution and format
- Figure and Table number consistent with text

Figures and Tables: *problems*

- Complicated figures or tables
- No or Inadequate description in the legends
- Mislabeling
- No markings (units, arrows, mol weight markers etc)
- No correspondence with text

General Goal and *Specific Aims*

- Begin with a statement of long term overall goal
- List the specific aims one by one concisely
- Not too many aims; 2-4 best
- Be ***specific, no general and ambiguous*** statements
- Be ***realistic, no unachievable aims; best if*** supported by preliminary results
- Aims should be logically linked and arranged accordingly; testing your hypothesis
- ***Link to experiments in Experimental Design***

highlight

Specific Aims

brief

- **Specific Aim #1**. Identification of downstream genes involved in ----. This purpose of this aim will test the hypothesis that------- Specifically,--- Yeast two-hybrid technique will be used to ----- Deletion analysis will be used to ----- *This study will be able to allow us to identify -----*

Title

Approaches

Hypothesis

Importance

Approaches

Rationale and hypothesis

Specific Aim 1. ***To determine the role of nucleosomes in the regulation of Igk locus rearrangement.*** Our preliminary results showed that the V(D)J recombinase could not recognize RSS targets if they were arranged into a nucleosome structure. We propose experiments to extend these observations by 1) determining what fraction of the Jk gene segments are in the nucleosomal structure in cells undergoing rearrangement as compared with non-lymphoid cells, 2) determining if nucleosomes are phased across the Jk locus, 3)determining whether nucleosome remodeling complex can alter the accessibility of the Jk cluster in native or reconstituted chromatin. This analysis will give us insight of the mechanism of Igk gene rearrangement.

Significance

Experimental Design(1)

- ***Do not write as Materials and Methods***
- ***Design the experiments to solve the problems*** posed in the specific aims in logical order
- Be ***realistic***, do not plan too many experiments or out of your expertise; manageable; focus!
- Be ***logical***; step-by-step leading to your goal

Experimental Design(2)

- Updated technologies; Do not re-invent the wheel
- Competence in using techniques proposed
- Evaluate the design critically; alternative approaches,
- ***Control! Control! Control!***
- A schematic diagram for the overall design will help the reviewers understand

Results Evaluation and Expectation

- State clearly why good results will be expected because of previous studies, your expertise, novel technology etc.
- Reasonable time table for achieving results
- Alternative interpretations critically evaluated
- How the results will support your hypothesis
- Perspectives

Significance

•State ***concisely and clearly the*** importance of your study for the advance of the field or to the benefit of society etc.

•List potential contributions in short sentences

•Innovations

Abstract

- ***First impression to the reviewers***

- **Summarize key points in the proposal:**
Importance and Rationale, Hypothesis, Aims, Approaches (brief), Innovation, Preliminary Results supporting the proposal, Expected Results, Significance

- ***Write it last, following the logic of the proposal***

- Give possible answers to the problems you want to study

- Do not use this section to write long background or discussions

Importance	Early onset breast cancer has been the unique and major health problem in Taiwan with annual increase-----
Outstanding problems	The etiology and mechanisms of breast cancer in young women is unknown. It has been suggested that carcinogenesis in utero may account for the early-onset of this cancer----------
Hypothesis	To test this hypothesis, we have used fed pregnant mice with fatty diet and found increase in incidence of mammary tumors in the new born mice.----------
Experimental approaches	In this proposal, we will examine the alteration of mammary gland differentiation pathway in the mammary tumors derived from the new born mice using microarray, SAGE (serial analysis of gene expression) as well real-time PCR techniques ----------
Summary Answers	This study of alterations in mammary gland differentiation in the mammary tumor of new born mice will allow us to identify the genetic and molecular mechanisms involved in the early-onset breast cancer.
Significance	This information could be useful for diagnosis/treatment of early-onset breast cancer in Taiwan----------

Title

•Choose title carefully—must reflect the central theme: ***selling point***

•Attractive and easy to understand

•Concise and ***Informative: no noninformative*** words like "A study of---"

•***Avoid acronyms***

References

- ***Should correspond to the text***
- ***Updated; don't miss the most relevant ones***
- Consistent format; numbering will help
- Title of article will help the reviewers

Budget

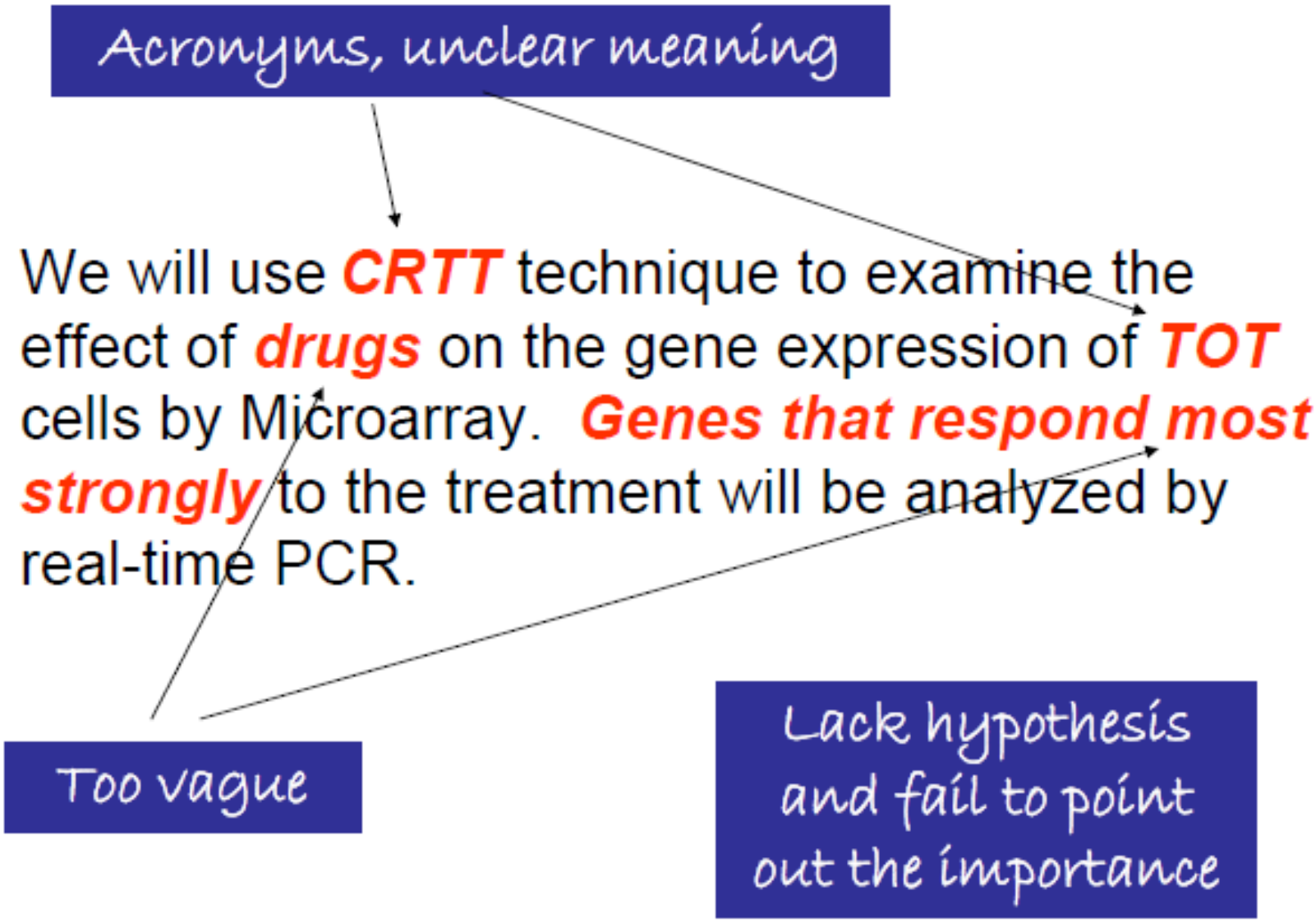
- Be reasonable
- State the ***rationale for unusually large*** request for fund
- State the ***justifications for purchasing the*** equipment needed
- Too many personnel to be hired is no good

Reasons Why Proposal is *Rejected(1)*

- Importance and rationale not clearly stated
- The problem is outmoded or trivial
- No specific aims! Aims too diffuse or general
- Similar experiments have already been done
- No evidence for competence to carry out the proposal
- Not realistic; overly ambitious plan

Reasons Why Proposal is *Rejected(2)*

- No experimental design! Or no correlation with the specific aims; poor design
- Too many factors or variables; No hypothesis; technique looking for problem
- No Controls
- Proposed study will not give useful information or new contributions
- Proposal depends on a key experiment with uncertain outcome


Acronyms, unclear meaning
We will use CRTT technique to examine the effect of drugs on the gene expression of TOT cells by Microarray. Genes that respond most strongly to the treatment will be analyzed by real-time PCR.
Too vague
Lack hypothesis and fail to point out the importance

Problems with Writing (1)

- Too long sentences and paragraphs
- Look out for long and dubious parentheses
- No logical connection among paragraphs
- Not explain specific terms clearly (reviewers probably are not in your field)
- ***acronyms and jargons***
- Misspelling; check with Word program
- Grammar; check with Word program
- Chaotic format